

EX. No. 14 – Simple Programming using REPEAT and WHILE

```
MySQL 8.0 Command Line Cli x + v
mysql> CREATE DATABASE IF NOT EXISTS LoopDB;
Query OK, 1 row affected, 1 warning (0.00 sec)

mysql> use LoopDB;
Database changed
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE SumEvenNumbers()
-> BEGIN
-> DECLARE n INT DEFAULT 2;
-> DECLARE total INT DEFAULT 0;
->
-> REPEAT
-> SET total = total + n;
-> SET n = n + 2;
-> UNTIL n > 10
-> END REPEAT;
-> SELECT total AS SumOfEvenNumbers;
-> END$$
ERROR 1304 (42000): PROCEDURE SumEvenNumbers already exists
mysql>
mysql> DELIMITER ;
mysql>
mysql> -- Call it
mysql> CALL SumEvenNumbers();
+-----+
| Sum_of_Even_Numbers |
+-----+
| 10100 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE FactorialWHILE(IN num INT)
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> DECLARE fact BIGINT DEFAULT 1;
->
-> WHILE i <= num DO
-> SET fact = fact * i;
-> SET i = i + 1;
-> END WHILE;
-> SELECT fact AS FactorialResult;
-> END$$
```

```
MySQL 8.0 Command Line Cli x + v
mysql>
mysql> -- Call it
mysql> CALL SumEvenNumbers();
+-----+
| Sum_of_Even_Numbers |
+-----+
| 10100 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE FactorialWHILE(IN num INT)
-> BEGIN
-> DECLARE i INT DEFAULT 1;
-> DECLARE fact BIGINT DEFAULT 1;
->
-> WHILE i <= num DO
-> SET fact = fact * i;
-> SET i = i + 1;
-> END WHILE;
-> SELECT fact AS FactorialResult;
-> END$$
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> -- Call it
mysql> CALL FactorialWHILE(5);
+-----+
| FactorialResult |
+-----+
| 120 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```